

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Overview

- The TCP/IP suite defines protocols mainly for the Application, Transport, and Network layers.
- The Data-Link and Physical layers are implemented by the underlying networks.
- These networks provide communication services to the upper layers.
- Both wired and wireless technologies operate at these lower layers.
- This module focuses on Data-Link Layer concepts and Wired Networks.

Topics Covered in Module 3

- 1 Introduction to Nodes and Links
- 2 Data Link Control (DLC)
- 3 Media Access Control (MAC)
- 4 Link-Layer Addressing and ARP
- 5 Ethernet and Wired LANs
- 6 Point-to-Point and Switched Networks
- 7 Connecting Devices

Learning Outcomes

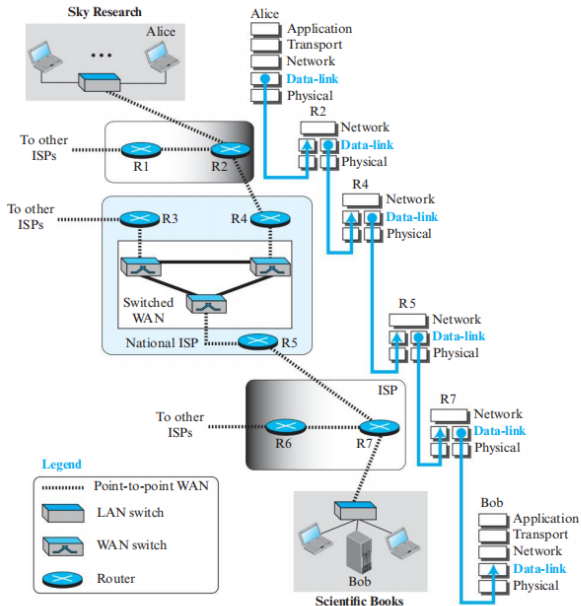
After completing this module, students will be able to:

- Explain the role of the Data-Link Layer.
- Describe framing, flow control, and error control mechanisms.
- Compare different MAC protocols.
- Understand link-layer addressing and ARP.
- Analyze Ethernet operation and evolution.
- Identify various wired networking technologies.
- Differentiate hubs, switches, and routers.

Host-to-Host Communication

- The Network Layer provides communication between source and destination hosts.
- Data is transferred in the form of **datagrams**.
- A datagram may travel through multiple networks before reaching its destination.
- These networks are interconnected using routers and switches.
- Therefore, successful delivery requires support from the Data-Link Layer at every hop.

Figure 5.1 *Communication at the data-link layer*



Logical Communication at the Data-Link Layer

Hop-by-Hop Communication

- Unlike the Network Layer, the Data-Link Layer provides **node-to-node** communication.
- Communication occurs separately over each individual link.
- In the path from Alice to Bob:
 - Alice $\leftrightarrow$ Router R2
 - Router R2 $\leftrightarrow$ Router R4
 - Router R4 $\leftrightarrow$ Router R5
 - Router R5 $\leftrightarrow$ Router R7
 - Router R7 $\leftrightarrow$ Bob
- Each router contains two Data-Link Layers:
 - One for the incoming network
 - One for the outgoing network
- Thus, end-to-end delivery is achieved through multiple hop-by-hop Data-Link communications.

Nodes and Links

Node-to-Node Communication

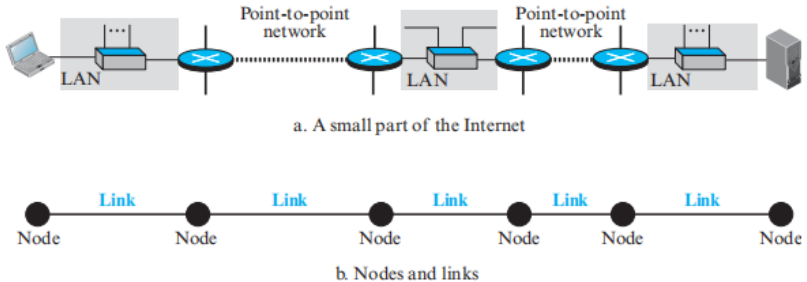
- Communication at the Data-Link Layer is **node-to-node**.
- Data traveling across the Internet passes through multiple networks and routers.
- Hosts and routers are collectively called **nodes**.
- The networks connecting nodes are called **links**.
- A complete path consists of alternating nodes and links.

Example Path

Source Host → Router 1 → Router 2 → Router 3 → Router 4 →
Destination Host

- First and last nodes are hosts.
- Intermediate nodes are routers.
- Links may be LANs or WANs.

Figure 5.2 *Nodes and Links*



Nodes and Links Example

(Source Host – LAN – Router – WAN – Router – LAN – Destination Host)

- Nodes represent hosts and routers.
- Links represent communication networks.
- Data is forwarded hop-by-hop across links.

Two Types of Links

1. Point-to-Point Link

- Dedicated connection between exactly two devices.
- Entire channel capacity belongs to the connected devices.
- No competition for transmission medium.
- Example: Traditional telephone connection.

2. Broadcast Link

- Shared communication medium.
- Multiple devices use the same channel.
- Requires access control mechanisms.
- Example: Wireless cellular communication.

Comparison of Link Types

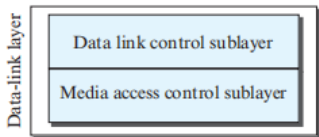
Feature	Point-to-Point	Broadcast
Medium Usage	Dedicated	Shared
Number of Devices	Two	Multiple
Channel Access Conflict	No	Yes
Efficiency	High	Depends on sharing
Examples	Telephone Link	Wi-Fi, Cellular Network

Two Sublayers of Data-Link Layer

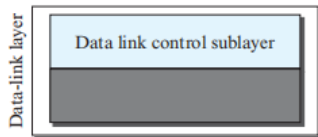
Data-Link Layer Structure

- To simplify design, the Data-Link Layer is divided into:
 - ① Data Link Control (DLC)
 - ② Media Access Control (MAC)
- Each sublayer performs specific functions.
- Together they provide reliable communication over a link.

Figure 5.3 *Dividing the data-link layer into two sublayers*



a. Data-link layer of a broadcast link



b. Data-link layer of a point-to-point link

Data Link Control (DLC)

Functions of DLC Sublayer

- Common to both point-to-point and broadcast links.
- Responsible for:
 - Framing
 - Flow Control
 - Error Detection
 - Error Control
- Ensures reliable transfer of frames between nodes.



Media Access Control (MAC)

Functions of MAC Sublayer

- Primarily used in broadcast networks.
- Controls access to the shared medium.
- Determines which device can transmit at a given time.
- Prevents or minimizes collisions.
- Used in Ethernet, Wi-Fi, and other shared networks.

Key Idea:

MAC manages "**who gets to use the channel**" while DLC manages "**how data is delivered reliably**".

Data Link Control (DLC)

Definition

- Data Link Control (DLC) is a sublayer of the Data-Link Layer.
- It manages communication between two adjacent nodes.
- DLC provides reliable node-to-node communication.
- It operates on both:
 - Point-to-Point Links
 - Broadcast Links

Goal: Ensure that data is delivered accurately and efficiently across a single link.

Major Functions of DLC

Data Link Control provides four major functions:

- 1 Framing
- 2 Flow Control
- 3 Error Control
- 4 Error Detection and Correction

Why DLC is Important?

The Physical Layer only transmits bits. DLC organizes, manages, and protects these bits during transmission.

Functions of DLC

Function	Purpose
Framing	Organizes bits into manageable units called frames.
Flow Control	Prevents a fast sender from overwhelming a slow receiver.
Error Control	Ensures reliable delivery of frames.
Error Detection	Detects transmission errors in frames.
Error Correction	Corrects or retransmits damaged frames.

This section discusses:

- ① Framing
 - Organizing bits into frames
- ② Flow Control
 - Regulating transmission rate
- ③ Error Control
 - Handling lost or damaged frames
- ④ Error Detection and Correction
 - Detecting and recovering from transmission errors

Definition

- The Physical Layer transmits a continuous stream of bits.
- The Data-Link Layer groups these bits into units called **frames**.
- Framing makes each message distinguishable from others.
- Frames are the basic units of transmission at the Data-Link Layer.

Key Idea

Framing organizes a stream of bits into manageable and identifiable units.

Why Do We Need Framing?

- Separates one message from another.
- Identifies sender and receiver.
- Supports flow control and error control.
- Makes retransmission efficient.

Analogy: Postal System

- Letter = Data
- Envelope = Frame
- Address on Envelope = Source and Destination Address

Role of Framing

- Each frame contains:

- Source Address
- Destination Address
- Data
- Control Information

““

- Destination address identifies the receiver.
- Source address helps acknowledgements and error recovery. ““

Why Not Use One Large Frame?

- A complete message could be sent as one frame.
- However, large frames are inefficient.

Problems with Large Frames

- Higher retransmission cost.
- Increased error recovery time.
- Poor flow control performance.

Example

If a 10 MB frame contains one bit error, the entire frame must be retransmitted.

Two Types of Frames

- ① Fixed-Size Frames
- ② Variable-Size Frames

Fixed-Size Framing

- Every frame has the same length.
- Frame boundaries are easily identified.
- Example: ATM Cells.



Variable-Size Framing

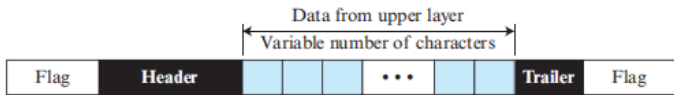
- Frame length can vary.
- Frame boundaries must be explicitly identified.
- Two approaches are used:
 - 1 Character-Oriented (Byte-Oriented) Framing
 - 2 Bit-Oriented Framing

Character-Oriented Framing

- Data is organized as 8-bit characters.
- Special flag bytes indicate:
 - Start of Frame
 - End of Frame
- Header and trailer are also multiples of 8 bits.



Figure 5.4 *A frame in a character-oriented protocol*



Byte Stuffing

- A flag pattern may appear inside the data.
- Receiver may incorrectly detect end of frame.
- Solution: Byte Stuffing.

Byte Stuffing

Insert an extra escape character (ESC) before any flag byte appearing in the data.

- Receiver removes ESC during processing.
- Actual flag remains distinguishable.

Handling Escape Characters

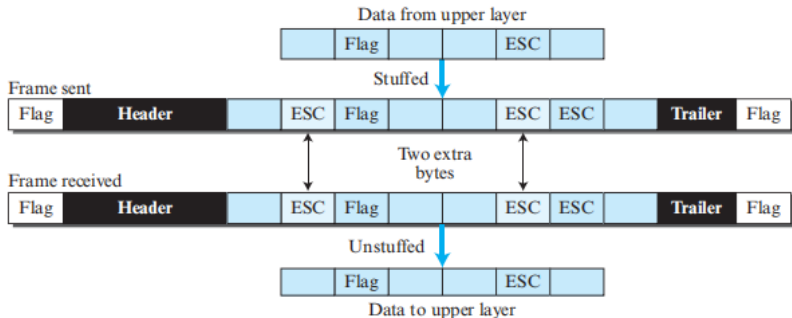
- What if ESC itself appears in data?
- Another ESC is inserted before it.

Rule

Flag $\rightarrow$ ESC + Flag

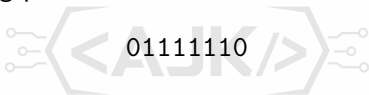
ESC $\rightarrow$ ESC + ESC

Figure 5.5 *Byte stuffing and unstuffing*



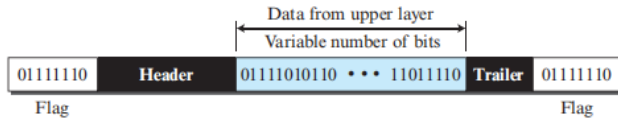
Bit-Oriented Framing

- Data is treated as a stream of bits.
- Uses a special flag pattern:



- Flag marks the beginning and end of a frame.
- Independent of character encoding.

Figure 5.6 *A frame in a bit-oriented protocol*



Bit Stuffing

- Flag pattern may accidentally appear in data.
- Solution: Bit Stuffing.

Bit Stuffing Rule

Whenever five consecutive 1s follow a 0, an extra 0 is inserted.

Example

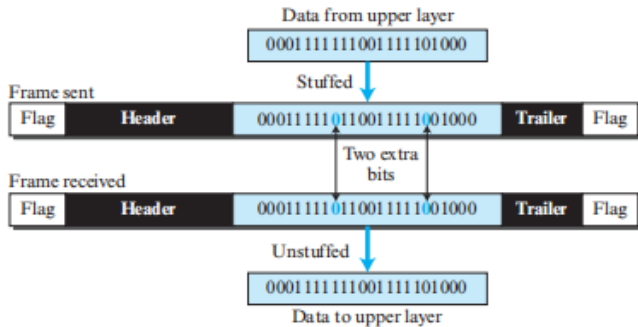
01111110 (Flag Pattern)

becomes

011111010

- Receiver removes stuffed 0.
- Prevents confusion with actual flags.

Figure 5.7 *Bit stuffing and unstuffing*



Bit Stuffing Example

- Sender inserts stuffed bits.
- Receiver detects and removes stuffed bits.
- Original data is restored correctly.

Character-Oriented vs Bit-Oriented Framing

Feature	Character-Oriented	Bit-Oriented
Unit	Bytes	Bits
Delimiter	Flag Byte	Flag Bit Pattern
Transparency Method	Byte Stuffing	Bit Stuffing
Supports Binary Data	Limited	Excellent
Modern Usage	Rare	Common

Character-Oriented vs Bit-Oriented Framing

Character-Oriented Framing

- Uses bytes as the basic unit.
- Frame delimiter is a flag byte.
- Uses Byte Stuffing.
- Suitable for text-based communication.

Bit-Oriented Framing

- Uses bits as the basic unit.
- Frame delimiter is the flag pattern 01111110.
- Uses Bit Stuffing.
- Suitable for modern data communication.

Responsibilities of DLC

- The Data Link Control (DLC) sublayer ensures reliable communication between adjacent nodes.
- Two important functions are:
 - 1 Flow Control
 - 2 Error Control
- These functions operate on a single link between two neighboring devices.

Definition

- Flow control regulates the rate of data transmission.
- It prevents a fast sender from overwhelming a slow receiver.
- The sender transmits only the amount of data that the receiver can handle.

At the Data-Link Layer

- Flow control is **node-to-node**.
- It operates across a single communication link.
- Acknowledgments are used to coordinate transmission.

Flow Control: Transport Layer vs Data-Link Layer

Feature	Transport Layer	Data-Link Layer
Communication Type	End-to-End	Node-to-Node
Scope	Host-to-Host	Adjacent Nodes
Purpose	Control overall data flow	Control flow across a link
Acknowledgment	End-to-End ACKs	Link-Level ACKs
Operation	Across entire network path	Across a single link

Key Point

Transport Layer flow control is **host-to-host**, whereas Data-Link Layer flow control is **node-to-node (link-by-link)**.

Error Control

Definition

- Error control ensures reliable frame delivery.
- It consists of:
 - Error Detection
 - Error Correction / Retransmission

Operation

- Receiver detects damaged or lost frames.
- Receiver informs the sender.
- Sender retransmits the affected frame.
- Each link independently checks frame integrity.

Key Point

Error control at the Data-Link Layer is node-to-node, whereas Transport Layer error control is end-to-end.

Error Detection and Correction

- Frames may become corrupted during transmission due to noise and interference.
- Data-link protocols usually detect errors and request retransmission.
- Some wireless protocols also perform error correction.
- Error handling relies on adding redundant bits to the transmitted data.

Goal

Detect or correct transmission errors before forwarding frames to the next node.

Types of Errors

- **Single-Bit Error**

- Only one bit changes from 0 to 1 or 1 to 0.

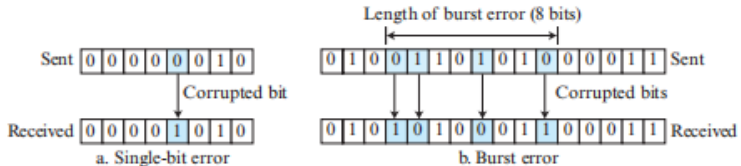
- **Burst Error**

- Two or more bits are altered.
- More common than single-bit errors.
- Noise usually affects multiple consecutive bits.

Observation

Burst errors are much more likely because noise duration is usually longer than one bit time.

Figure 5.8 *Single-bit and burst error*



Redundancy and Error Handling

- Error detection and correction require extra bits called **redundancy bits**.
- Redundant bits are:
 - Added by the sender.
 - Removed by the receiver.
- These bits help identify corrupted data during transmission.

Key Idea

No error detection or correction is possible without adding redundancy.

Error Detection vs Error Correction

Error Detection

- Determines whether an error occurred.
- Output is simply:
 - Error Present
 - No Error
- Simpler process.

Error Correction

- Identifies:
 - Number of errors
 - Location of errors
- More complex.
- Requires additional redundancy.

Important

Error correction is significantly harder than error detection because the receiver must locate and repair corrupted bits.

Coding for Error Detection

- Redundancy is introduced using coding techniques.
- Sender generates redundant bits based on data bits.
- Receiver verifies the relationship between data and redundant bits.

Two Main Coding Techniques

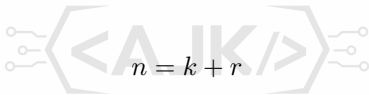
- 1 Block Coding
- 2 Convolution Coding

Note

Computer networks primarily use Block Coding for error detection.

Block Coding

- Message is divided into blocks of **k bits**.
- Each block is called a **Dataword**.
- Add **r redundant bits**.
- Total length becomes:



A diagram illustrating the relationship between a dataword and a codeword. It features a large, light gray stylized letter 'A' in the background. In the foreground, there is a hexagonal shape with circuit-like lines extending from its left and right sides. Inside the hexagon, the equation $n = k + r$ is written in a light gray font.

$$n = k + r$$

- Resulting block of n bits is called a **Codeword**.

Example

Dataword = 4 bits

Redundant bits = 3 bits

Codeword = 7 bits

Datawords and Codewords

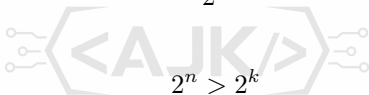
- Number of possible datawords:

$$2^k$$

- Number of possible codewords:

$$2^n$$

- Since $n > k$,


$$2^n > 2^k$$

- Many possible codewords remain unused.

Important

Unused codewords are called **Invalid** or **Illegal Codewords**.

Role of Invalid Codewords

- Valid codewords correspond to actual datawords.
- Invalid codewords are never generated by the sender.
- If transmission errors convert a valid codeword into an invalid codeword:



Error Detection Principle

Presence of invalid codewords allows receivers to identify transmission errors.

Conditions for Error Detection

For block coding to detect errors:

- 1 Receiver must know all valid codewords.
- 2 Error must transform a valid codeword into an invalid codeword.

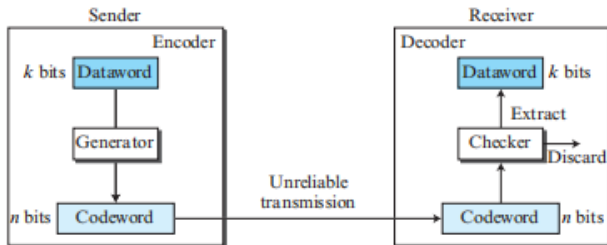
Limitation

If a corrupted codeword accidentally becomes another valid codeword, the error remains undetected.

Block Coding Process

- ❶ Sender converts datawords into codewords.
- ❷ Codewords are transmitted through the channel.
- ❸ Noise may alter bits during transmission.
- ❹ Receiver checks whether received codeword is valid.
 - Valid Codeword → Accept Data
 - Invalid Codeword → Reject Data

Figure 5.9 *Process of error detection in block coding*



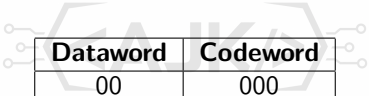
Summary: Error Detection and Correction

- Errors occur because of noise and interference.
- Errors may be:
 - Single-bit errors
 - Burst errors
- Redundancy enables detection and correction.
- Block coding adds redundant bits to create codewords.
- Invalid codewords help detect transmission errors.
- Error detection is simpler than error correction.

Example: Error Detection Using Block Coding

Given:

- Dataword size: $k = 2$
- Codeword size: $n = 3$
- Four valid codewords exist for the four possible datawords.



Dataword	Codeword
00	000
01	011
10	101
11	110

Scenario

Sender transmits dataword 01 as codeword 011.

Example 5.1 – Case 1: No Error

Transmitted Codeword:

011

Received Codeword:

011

- Received codeword matches a valid codeword.
- Receiver accepts the codeword.
- Extracted dataword:

01

Result

No transmission error occurred.

Example 5.1 – Case 2: Detectable Error

Transmitted Codeword:

011

Received Codeword:

111

- Leftmost bit is corrupted.
- Codeword 111 is not in the list of valid codewords.
- Receiver identifies it as an invalid codeword.
- Frame is discarded.

Result

Error detected successfully.

Example 5.1 – Case 3: Undetectable Error

Transmitted Codeword:

011

Received Codeword:

000

- Two bits are corrupted.
- Received codeword 000 is still a valid codeword.
- Receiver incorrectly interprets it as:

00

Result

Error is NOT detected because corrupted bits transformed one valid codeword into another valid codeword.

Key Observation

Important Conclusion

An error-detecting code can detect only the types of errors for which it is designed.

- If a valid codeword becomes an invalid codeword:
 - Error is detected.
- If a valid codeword becomes another valid codeword:
 - Error remains undetected.

Valid $\rightarrow$ Invalid = Detectable Error

Valid $\rightarrow$ Valid = Undetectable Error

Hamming Distance

- Hamming Distance is a fundamental concept in error detection and correction.
- It represents the number of bit positions in which two codewords differ.

$$d(x, y) = \text{Number of differing bits between } x \text{ and } y$$

Importance

The Hamming distance between the transmitted and received codeword equals the number of corrupted bits during transmission.

Calculating Hamming Distance

- Hamming distance can be found using the XOR operation.
- Perform:

$$x \oplus y$$

- Count the number of 1s in the result.

Formula

$$d(x, y) = \text{Number of 1s in } (x \oplus y)$$

Example

$$00000 \oplus 01101 = 01101$$

Number of 1s = 3

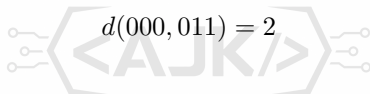
$$d(00000, 01101) = 3$$

Example 5.2 – Hamming Distance

Example 1

$$000 \oplus 011 = 011$$

Number of 1s = 2


$$d(000, 011) = 2$$

Example 2

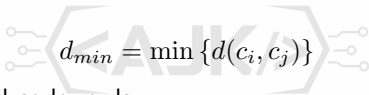
$$10101 \oplus 11110 = 01011$$

Number of 1s = 3

$$d(10101, 11110) = 3$$

Minimum Hamming Distance

- In a coding scheme, many valid codewords exist.
- The **minimum Hamming distance** (d_{min}) is the smallest distance between any two valid codewords.


$$d_{min} = \min \{d(c_i, c_j)\}$$

where c_i and c_j are valid codewords.

Significance

The larger the minimum Hamming distance, the better the error detection capability.

Error Detection Requirement

Suppose a code must detect up to s errors.

- If s bits are corrupted,

$$d(\text{sent}, \text{received}) = s$$

- To guarantee detection, valid codewords must be farther apart than s .

Rule

To detect up to s errors,

$$d_{\min} = s + 1$$

Why $d_{min} = s + 1$?

- Assume the transmitted codeword is x .
- Up to s bit errors can move the received codeword away from x .
- If another valid codeword exists within distance s , the receiver may mistake it for the transmitted codeword.

Guarantee

If all valid codewords are separated by at least

$$s + 1$$

bits, any error involving s or fewer bits will always be detected.

Geometric Interpretation

- Consider the transmitted codeword as the center of a circle.
- Radius of the circle = s .
- All codewords produced by up to s errors lie inside the circle.
- No other valid codeword should lie inside this region.

Condition

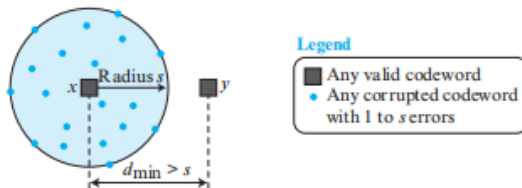
Valid codewords must be outside the error circle.

$$d_{min} > s$$

Therefore,

$$d_{min} = s + 1$$

Figure 5.10 *Geometric concept explaining $d_{\min}$ in error detection*



Summary

- Hamming Distance = Number of differing bits between two codewords.
- Computed using XOR and counting 1s.
- Minimum Hamming Distance (d_{min}) is the smallest distance among valid codewords.
- Larger d_{min} provides better error detection capability.

Key Formula

To guarantee detection of up to s errors:

$$d_{min} = s + 1$$

Example 5.3: Minimum Hamming Distance = 2

Recall the code from Table 5.1:

Dataword	Codeword
00	000
01	011
10	101
11	110

- Minimum Hamming Distance:

$$d_{min} = 2$$

- Using the rule:

$$d_{min} = s + 1$$

$$2 = s + 1$$

$$s = 1$$

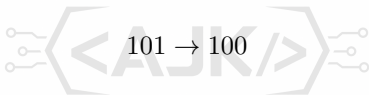
Example 5.3: Single Error Detection

Assume codeword:

101

is transmitted.

Suppose one bit changes during transmission:



- Received codeword = 100
- 100 is not a valid codeword.
- Receiver detects the error.

Observation

Any single-bit error transforms a valid codeword into an invalid codeword.

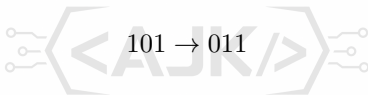
Example 5.3: Two Errors May Go Undetected

Assume codeword:

101

is transmitted.

Suppose two bits are corrupted:



- Received codeword = 011
- 011 is also a valid codeword.
- Receiver accepts the codeword incorrectly.

Result

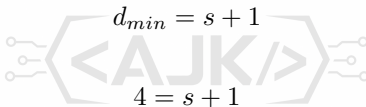
Two-bit errors may not be detected when $d_{min} = 2$.

Example 5.4: Hamming Distance = 4

Given:

$$d_{min} = 4$$

Using the error detection rule:


$$d_{min} = s + 1$$
$$4 = s + 1$$

$$s = 3$$

Result

A code with minimum Hamming distance 4 can detect up to 3 errors.

Error Detection Capability

Minimum Hamming Distance	Errors Detectable
2	1 Error
3	2 Errors
4	3 Errors
5	4 Errors

General Rule

$$d_{min} = s + 1$$

To guarantee detection of up to s errors.